

Neural Networks

A neural network is a machine learning model composed of interconnected layers of computational units called neurons.

A basic neural network contains an input layer, one or more hidden layers, and an output layer.

The activation function introduces non-linearity into a neural network. ReLU is commonly used in hidden layers.

During training, backpropagation calculates gradients of the loss with respect to model parameters.

An optimizer updates weights using the calculated gradients. Adam is a widely used optimization algorithm.

The learning rate controls the size of parameter updates during training.

A convolutional neural network, or CNN, is especially useful for processing images because convolution layers can learn spatial features.